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EE1030/EE2100: Matrix Theory

Lecture 16: Orthogonal Complement of a Subspace

Topics :

1. Projection onto Subspace
 2. Orthogonal Complement of a Subspace
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Orthogonal Complement of a Subspace

Def: Let $W \subseteq \mathbb{R}^n$ be a subspace. The orthogonal complement of W , denoted by $W^\perp$ is the collection of vectors that are orthogonal to every vector in W .

at a later point

$$W^\perp = \{ \vec{v} : \vec{v} \perp \vec{w} \forall \vec{w} \in W \}$$

$W = \text{Col space}(A)$

Example: $W = \text{Span}(\vec{e}_1, \vec{e}_2) \subseteq \mathbb{R}^3$ (xy plane) $\rightarrow \dim(W) = 2$

$W^\perp = \text{Null space}(A)$

$W^\perp = \text{Collection of all vectors of form } \alpha \vec{e}_3 \rightarrow \dim(W^\perp) = 1$

$$\left. \begin{array}{l} \dim(W) = 2 \\ \dim(W^\perp) = 1 \end{array} \right\} \dim(W) + \dim(W^\perp) = 3$$

Note: if $v \notin W \nRightarrow v \in W^\perp$

Example 2: $W = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\} \quad \dim(W) = 1$

$$W^\perp = \{ \vec{v} : \vec{v}^T \vec{b} = 0 \} \Rightarrow v_1 + 2v_2 + 3v_3 = 0 \Rightarrow \text{fix } v_1, v_2 \text{ then}$$

$$\text{any } \vec{v} \text{ of the form } \begin{bmatrix} v_1 \\ v_2 \\ -\frac{1}{3}(v_1 + 2v_2) \end{bmatrix} = v_1 \underbrace{\begin{bmatrix} 1 \\ 0 \\ -1/3 \end{bmatrix}}_{\vec{b}_2} + v_2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ -2/3 \end{bmatrix}}_{\vec{b}_3}$$

$\vec{v} = \text{any linear comb of } \vec{b}_2 \text{ \& } \vec{b}_3$

$= \text{Span}(\vec{b}_2, \vec{b}_3) \Rightarrow W^\perp \text{ is a subspace}$
 $\text{ \& basis of } W^\perp = (\vec{b}_2, \vec{b}_3)$

Orthogonal Complement of a Subspace

The only vec in W and $W^\perp$ is $\vec{0}$ $W \cap W^\perp = \{\vec{0}\} \Rightarrow$ for $\vec{v}$ to be in W & $W^\perp$, it must satisfy $\langle \vec{v}, \vec{v} \rangle = 0 \Rightarrow \vec{v} = \vec{0}$

$W^\perp$ is a Subspace: a) $\vec{0} \in W^\perp$ (from above)

b) Closed under vec. addition if $\vec{u}, \vec{v} \in W^\perp$

$$\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle = 0$$

where $\vec{w} \in W$

$$\Rightarrow \vec{u} + \vec{v} \perp \vec{w} \Rightarrow \vec{u} + \vec{v} \in W^\perp$$

c) Closed under scalar mul (if $\vec{v} \in W^\perp$, then $\alpha \vec{v} \in W^\perp \forall \alpha$)

$$\langle \alpha \vec{v}, \vec{w} \rangle = \alpha \langle \vec{v}, \vec{w} \rangle = 0$$

$$\Rightarrow \alpha \vec{v} \in W^\perp$$

$$\text{if } \vec{v} \notin W, \vec{w} = \text{Proj}_W \vec{v} \in W \text{ and } \vec{r} = \vec{v} - \vec{w} \in W^\perp \Rightarrow \vec{v} = \underbrace{\vec{w}}_{\in W} + \underbrace{\vec{r}}_{\in W^\perp}$$

Orthogonal Complement of a Subspace

Any $\vec{v}$ can be written uniquely as $\vec{w} + \vec{r}$ where $\vec{w} \in W$ and $\vec{r} \in W^\perp$

Proof for Uniqueness:-

$$\text{Let } \begin{cases} \vec{v} = \vec{w}_1 + \vec{r}_1 & \& \\ = \vec{w}_2 + \vec{r}_2 \end{cases} \quad \text{where } \vec{w}_1, \vec{w}_2 \in W \text{ and } \vec{r}_1, \vec{r}_2 \in W^\perp$$

$$\rightarrow \vec{w}_1 + \vec{r}_1 = \vec{w}_2 + \vec{r}_2$$

$$\Rightarrow \underbrace{\vec{w}_1 - \vec{w}_2}_{\in W} = \underbrace{\vec{r}_2 - \vec{r}_1}_{\in W^\perp}$$

only possible if $\vec{w}_1 - \vec{w}_2 = \vec{0}$ & $\vec{r}_1 - \vec{r}_2 = \vec{0}$

$$\Rightarrow \vec{w}_1 = \vec{w}_2 \text{ and } \vec{r}_1 = \vec{r}_2$$

$\dim(W) + \dim(W^\perp) = \dim(\mathbb{R}^n) = n$. (if $W \subseteq \mathbb{R}^n$)

Proof:- Let $\vec{b}_1, \dots, \vec{b}_m$ be the basis of W

$\vec{b}_1, \dots, \vec{b}_m \in \mathbb{R}^n$ & are linearly independent $\Rightarrow (\vec{b}_1, \dots, \vec{b}_m, \underbrace{\vec{b}_{m+1}, \dots, \vec{b}_n}_{\text{add these vectors}})$
so that overall collection is basis of $\mathbb{R}^n$

Orthogonal Complement of a Subspace

Apply gram Schmidt to the collection $(\vec{b}_1, \dots, \vec{b}_m, \vec{b}_{m+1}, \dots, \vec{b}_n)$

$$\Downarrow$$

$$(\vec{\phi}_1, \dots, \vec{\phi}_m, \vec{\phi}_{m+1}, \dots, \vec{\phi}_n)$$

orthogonal basis
of W

and $\vec{\phi}_i \perp W$ for $i = m+1$ to n

all vectors $\vec{v} \in \mathbb{R}^n$ that are $\perp W \Rightarrow \vec{v} = \sum_{i=1}^n \alpha_i \vec{\phi}_i$, $\alpha_i = 0$
 $\forall i = 1 \dots m$

$\Rightarrow \vec{v}$ is a linear comb of
 $(\vec{\phi}_{m+1}, \dots, \vec{\phi}_n)$

$\Rightarrow (\vec{\phi}_{m+1}, \dots, \vec{\phi}_n)$ is basis of $W^\perp$
and $\dim(W^\perp) = n - m$.